

EduManage - Project Development Plan

PROJECT OVERVIEW

EduManage ek School Management SaaS hai jo MERN stack par build hoga. Isme 4 type ke users honge: Super Admin, School Admin, Teacher, aur Parent/Student. Project ko 4 phases mein divide kiya gaya hai.

PHASE 1 - Foundation aur Authentication (Week 1-2)

Is phase mein poora project ka base tayyar hoga. Pehle folder structure banao. Backend mein config, controllers, middleware, models, routes, aur services ke folders honge. Frontend mein pages, components, aur store ke folders honge.

Backend setup mein yeh karna hai. Node.js project initialize karo. Express server setup karo. MongoDB connection lagao Mongoose ke zariye. Environment variables ka .env file banao jisme database URL, JWT secret, aur Cloudinary keys hongi.

Database models is phase mein banana hai. User model banao jisme name, email, password, role, aur schoolId ka field hoga. School model banao jisme name, address, logo, plan, aur isApproved ka field hoga.

Authentication system is phase mein complete hoga. Register API banao. Login API banao. JWT access token aur refresh token ka system lagao. Password hashing bcrypt se karo. Role-based middleware likho taake har route pe check ho ke kon sa user access kar sakta hai.

Frontend mein is phase mein sirf Login aur Register ka page banana hai. Redux Toolkit setup karo state management ke liye. Axios instance banao jo har request ke saath token bheje.

Phase 1 complete hone ke baad tumhare paas working login system hoga jisme 4 alag roles honge.

PHASE 2 - Core Features (Week 3-5)

Is phase mein asli kaam hoga. Teen cheezein build hongi: School Management, Student aur Teacher Management, aur Attendance System.

School Management mein Super Admin ka dashboard banega. Woh nayi schools ki requests dekh sakta hai. Approve ya reject kar sakta hai. Sari active schools ki list dekh sakta hai.

School Admin ka flow is phase mein hoga. School Admin apni school mein teachers add karega. Har teacher ko ek ya zyada classes assign karega. Students add karega aur unhe classes mein enroll karega. Class model mein class ka naam, schoolId, teacherId, aur subjects ki list hogi.

Student model mein name, rollNo, photo, schoolId, classId, aur parentId hoga. Parent ka account bhi is phase mein banega jo sirf apne bachay ki info dekh sakta hai.

Attendance System is phase ka important hissa hai. Teacher roz apni class ki attendance lega. Ek API banega jo ek saath poori class ki attendance save kare. Attendance model mein studentId, classId, date, status (present ya absent), aur markedBy hoga. School Admin attendance ki report dekh sakta hai jisme date range filter hoga.

Frontend mein is phase mein yeh pages banenge. School Admin ka dashboard. Teacher add karne ka form. Student add karne ka form. Attendance lene ka page jisme ek list hogi aur har student ke saath present ya absent ka toggle hoga.

PHASE 3 - Fees, Results, aur Notifications (Week 6-8)

Is phase mein paise ka hisaab aur results ka system banega.

Fees Management mein School Admin har student ki monthly fees set karega. Fees model mein studentId, schoolId, month, year, amount, status (paid ya unpaid), aur paidAt hoga. Admin ek click mein unpaid fees ki list dekh sakta hai. Fees paid mark karne par automatically parent ko notification jayegi.

Results System mein teacher har subject ke marks enter karega. Result model mein studentId, classId, subject, marks, totalMarks, aur exam type (mid ya final) hoga. Parent apne bachay ka result dekh sakta hai. PDF result card generate hoga PDFKit se.

Notifications System poore project ko connect karega. Jab fees unpaid ho to reminder jayega. Jab bacha absent ho to parent ko alert jayega. Jab result publish ho to notification jayegi. Notification model mein userId, schoolId, type, message, isRead, aur createdAt hoga. Socket.io real-time notifications ke liye use hoga taake browser mein turant dikhe bina page refresh kiye.

Email notifications Nodemailer se hongii. Cloudinary se school ka logo aur student ki photo upload hogi.

Frontend mein is phase mein Fees Management ka page banega. Results enter karne ka form banega. Parent portal banega jisme bacha ki attendance, fees, aur result ek jagah dikhe. Notification bell icon banega header mein jo real-time update ho.

PHASE 4 - Subscription, Polish, aur Deployment (Week 9-10)

Is phase mein project ko production-ready banaya jayega.

Subscription System mein Stripe payment integrate hoga. Do plans honge: Basic aur Pro. Basic plan mein 200 students tak ki limit hogi aur fee 2000 PKR per month hogi. Pro plan unlimited students ke liye hoga aur fee 5000 PKR per month hogi. School Admin dashboard mein plan upgrade karne ka option hoga. Payment successful hone par school ka plan automatically update hoga.

Security hardening is phase mein hoga. Rate limiting lagao taake koi ek IP bahut zyada requests na bhej sake. Helmet.js lagao HTTP headers secure karne ke liye. Input validation express-validator se karo. CORS properly configure karo.

Testing is phase mein hoga. Har API ko Postman se test karo. Sab edge cases cover karo jaise wrong password, duplicate email, unauthorized access.

Deployment mein Backend Render ya Railway par deploy hoga. Frontend Vercel par deploy hoga. MongoDB Atlas use hoga production database ke liye. Environment variables production server par set hongi. Custom domain connect hoga agar available ho.

Portfolio preparation is phase ka aakhri kaam hai. README file likho jisme project ka description, features, installation steps, aur live link ho. Screenshots lo har role ka dashboard. Loom ya OBS se ek demo video banao 3-5 minute ka. GitHub par code clean kar ke push karo meaningful commit messages ke saath.

TECH STACK SUMMARY

Backend mein Node.js, Express, MongoDB, Mongoose, JWT, bcrypt, Socket.io, Nodemailer, PDFKit, aur Cloudinary use hoga. Frontend mein React, Redux Toolkit, Axios, React Router, aur Leaflet use hoga. Payment ke liye Stripe hoga. Deployment ke liye Vercel aur Render honge.

TOTAL TIMELINE: 10 weeks

Phase 1: Week 1 to 2 - Foundation aur Auth Phase 2: Week 3 to 5 - Core Features Phase 3: Week 6 to 8 - Fees, Results, Notifications Phase 4: Week 9 to 10 - Subscription aur Deployment

Phase 1 se shuru karte hain? Main folder structure aur pehla code likh deta hun.

